

Lab 4: Inserting Initial Data into the Database

Scenario:

The store manager wants to add initial product categories and products to the system.

Objective:

Use EF Core to insert records using `AddAsync` and `SaveChangesAsync`.

Source code

Category.cs

```
namespace EFCoreStore.Models
{
    public class Category
    {
        public int CategoryId { get; set; }
        public string Name { get; set; }
        public List<Product> Products { get; set; } = new();
    }
}
```

Product.cs

```
namespace EFCoreStore.Models
{
    public class Product
    {
        public int ProductId { get; set; }
        public string Name { get; set; }
        public decimal Price { get; set; }
        public int CategoryId { get; set; }
        public Category Category { get; set; }
    }
}
```

```
}
```

AppDbContext.cs

```
using Microsoft.EntityFrameworkCore;
using EFCoreStore.Models;
namespace EFCoreStore.Data
{
    public class AppDbContext : DbContext
    {
        public DbSet<Category> Categories { get; set; }

        public DbSet<Product> Products { get; set; }

        protected override void OnConfiguring(DbContextOptionsBuilder optionsBuilder)
        {
            optionsBuilder.UseSqlServer(

"Server=YOUR_SERVER_NAME;Database=StoreDB;Trusted_Connection=True;TrustServerCertificate=True;");
        }
    }
}
```

Program.cs

```
using EFCoreStore.Data;
using EFCoreStore.Models;
using var context = new AppDbContext();
var electronics = new Category
{
    Name = "Electronics"
};
var groceries = new Category
{
```

```
        Name = "Groceries"
    };

await context.Categories.AddRangeAsync(electronics, groceries);

var product1 = new Product
{
    Name = "Laptop",
    Price = 75000,
    Category = electronics
};

var product2 = new Product
{
    Name = "Rice Bag",
    Price = 1200,
    Category = groceries
};

await context.Products.AddRangeAsync(product1, product2);

await context.SaveChangesAsync();

Console.WriteLine("Data inserted successfully.");
```

SQL

SELECT * FROM Categories;

CategoryId	Name
1	Electronics
2	Groceries

SELECT * FROM Products;

ProductId	Name	Price	CategoryId
1	Laptop	75000	1
2	Rice Bag	1200	2

SELECT

p.ProductId,

p.Name,

p.Price,

c.Name AS Category

FROM Products p

INNER JOIN Categories c

ON p.CategoryId = c.CategoryId;

ProductId	Product	Price	Category
1	Laptop	75000	Electronics
2	Rice Bag	1200	Groceries

Data inserted successfully.

```
PS C:\EFCoreStore> dotnet run
```

Data inserted successfully.

```
PS C:\EFCoreStore>
```